

UNIT-3

(Crystal Structure)

Why Study Solid State Physics?

Why is there Interest in Solid State Physics?

Electronic Properties

- » Transistors (the heart of EVERYTHING electronic)

Electron/Photon Interactions

- » Laser Diodes, Photodiodes and CCDs

Electron/Phonon Interactions

- » Piezoelectric materials

Electron Spin/Charge Interactions

- » Spintronics, Quantum Computation

Highly-Correlated Electronic Systems

- » Superconductors



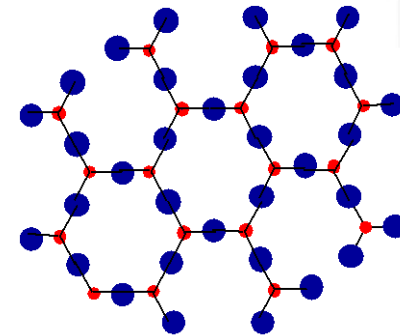
Structures of Metals

- How do atoms assemble into solid structures?
- How does the density of a material depend on its structure?
- When do material properties vary with the sample (i.e., part) orientation?

MATERIALS AND PACKING

Crystalline materials...

- atoms pack in periodic, 3D arrays
- typical of:
 - metals
 - many ceramics
 - some polymers



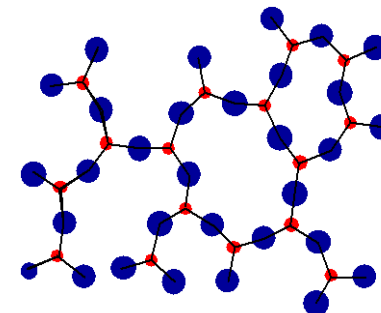
crystalline SiO₂

Adapted from Fig. 3.18(a),
Callister 6e.

• Si • Oxygen

Noncrystalline materials...

- atoms have no periodic packing
- occurs for:
 - complex structures
 - rapid cooling



noncrystalline SiO₂

Adapted from Fig. 3.18(b),
Callister 6e.

"Amorphous" = Noncrystalline

Ideal Crystal

- An ideal crystal is a **periodic array** of structural units, such as atoms or molecules.
- It can be constructed by the infinite repetition of these identical structural units in space.
- Structure can be described in terms of a **lattice**, with a group of atoms attached to each lattice point. The group of atoms is the **basis**.

Bravais Lattice

- An infinite array of discrete points with an arrangement and orientation that appears **exactly the same**, from any of the points the array is viewed from.
- A three dimensional Bravais lattice consists of all points with position vectors **R** that can be written as a linear combination of *primitive vectors*.

$$\mathbf{R} = n_1 \mathbf{a}_1 + n_2 \mathbf{a}_2 + n_3 \mathbf{a}_3$$

The expansion coefficients n_1, n_2, n_3 must be **integers**. The Vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ are primitive vectors.

Crystal Structure

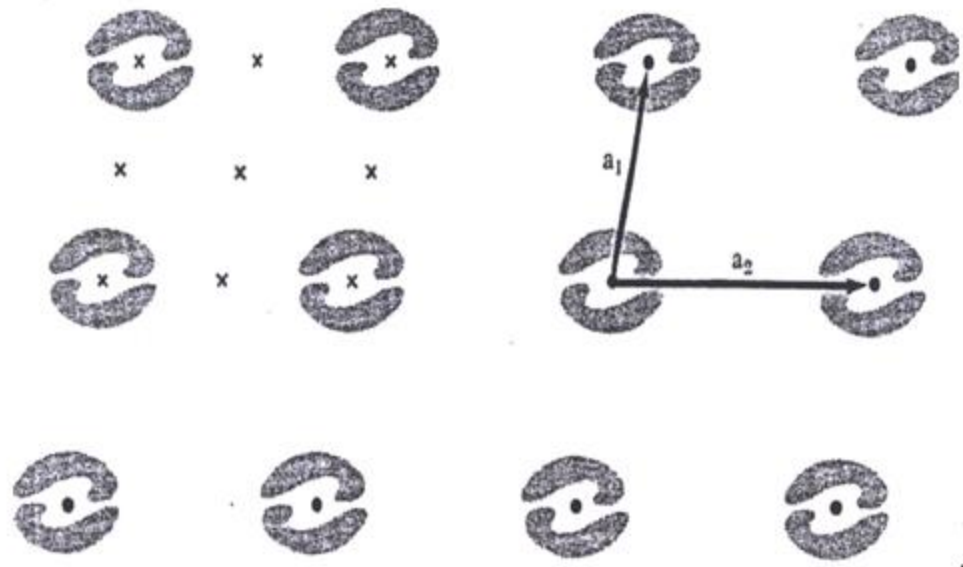


Figure 3 Similar to Fig. 2, but with protein molecules associated in pairs. The crystal translation vectors are a_1 and a_2 . A rotation of π radians about any point marked $\times$ will carry the crystal into itself. This occurs also for equivalent points in other cells. but we have marked the points $\times$ only

Crystal structure: basis

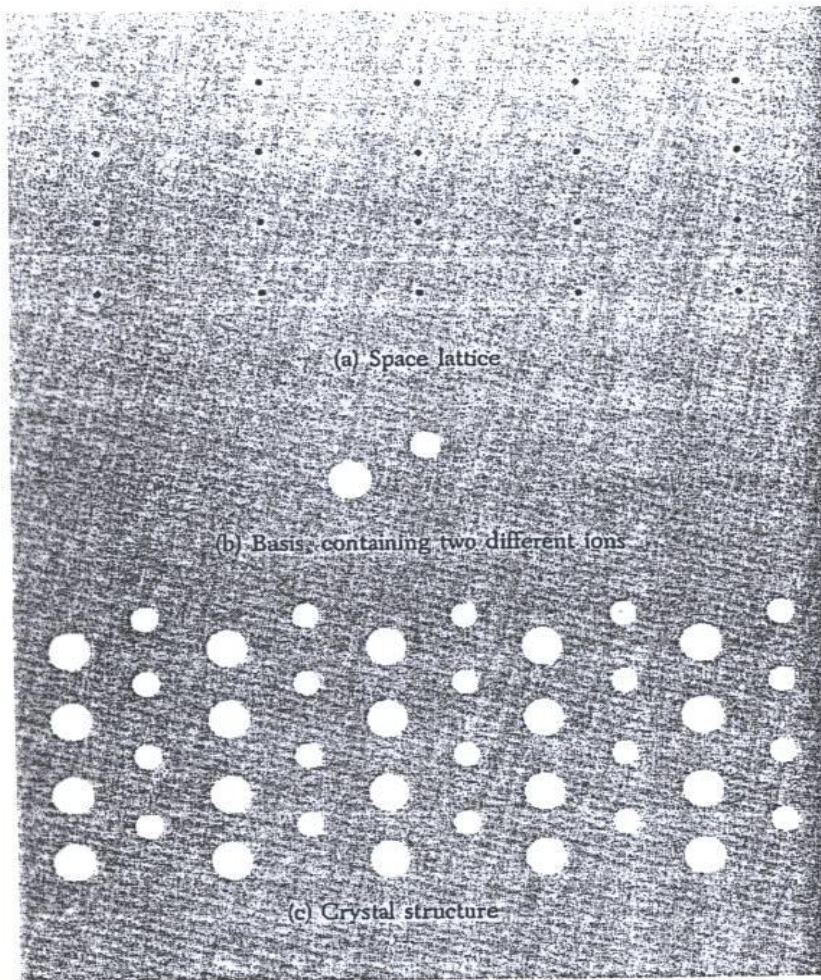


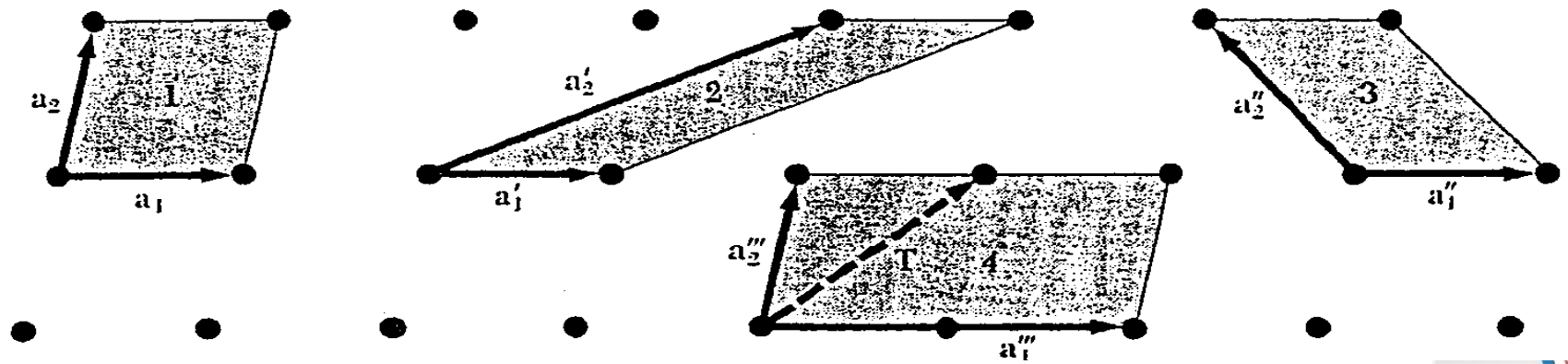
Figure 4 The crystal structure is formed by the addition of the basis (b) to every lattice point of the lattice (a). By looking at (c), you can recognize the basis and then you can abstract the space lattice. It does not matter where the basis is put in relation to a lattice point.

Translation Vector $\mathbf{T}$

$$\mathbf{T} = u_1 \mathbf{a}_1 + u_2 \mathbf{a}_2 + u_3 \mathbf{a}_3$$

Any two lattice points are connected by a vector of this form.

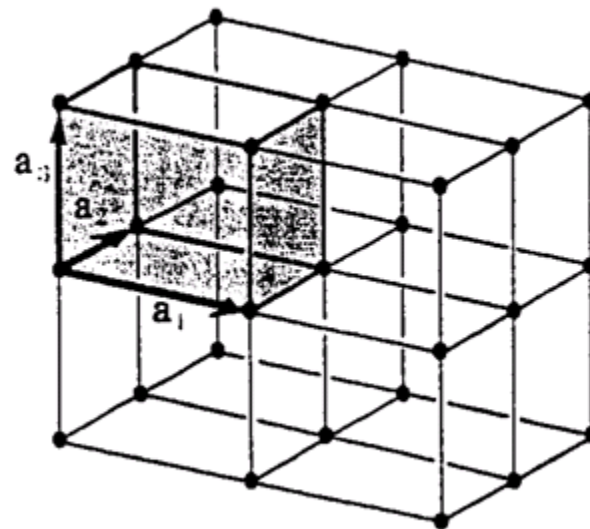
Translation(a_1, a_2), Nontranslation Vectors(a_1''', a_2''')



Primitive Unit Cell

- A **primitive cell** or **primitive unit cell** is a volume of space that when translated through all the vectors in a Bravais lattice just fills all of space without either overlapping itself or leaving voids.
- A primitive cell must contain precisely one lattice point.

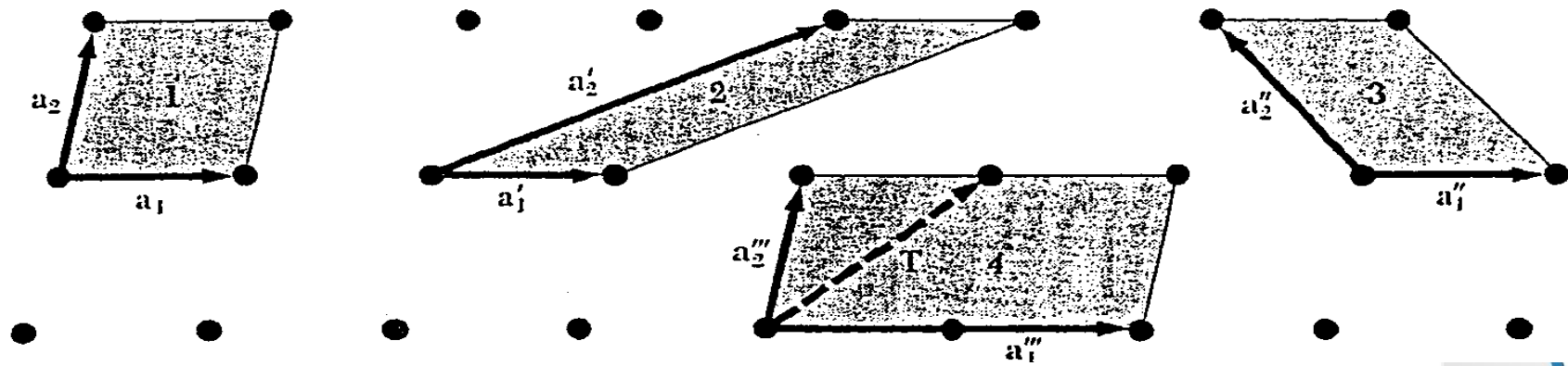
Primitive cell of a space lattice in three dimensions.



Two Dimensional Lattices

- There is an unlimited number of possible lattices, since there is no restriction on the lengths of the lattice translation vectors or on the angle between them. An **oblique lattice** has arbitrary **$\mathbf{a}_1$** and **$\mathbf{a}_2$** and is invariant only under rotation of π and 2π about any lattice point.

Oblique lattice: invariant only under rotation of π and 2π



Two Dimensional Lattices

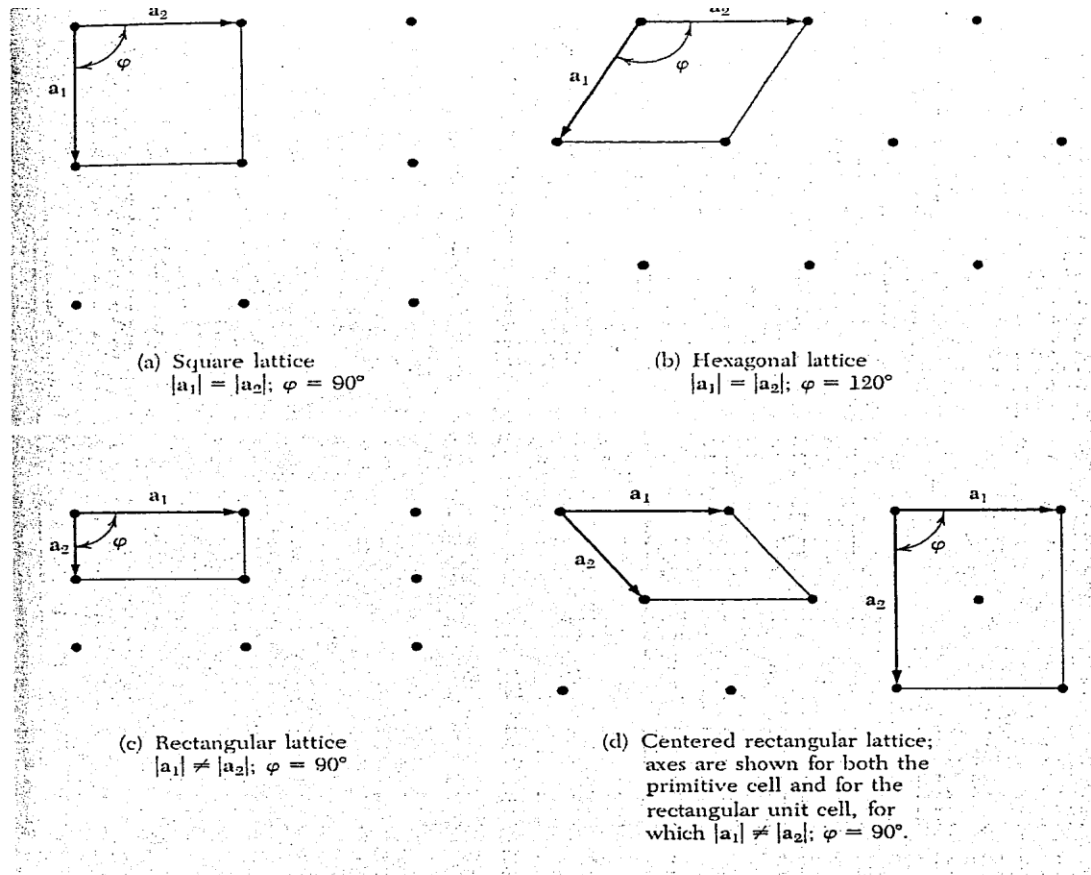
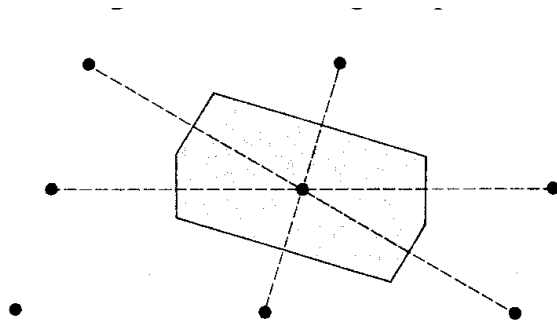


Figure 9

Three Dimensional Lattice Types

System	Number of lattices	Restrictions on conventional cell axes and angles
Triclinic	1	$a_1 \neq a_2 \neq a_3$ $\alpha \neq \beta \neq \gamma$
Monoclinic	2	$a_1 \neq a_2 \neq a_3$ $\alpha = \gamma = 90^\circ \neq \beta$
Orthorhombic	4	$a_1 \neq a_2 \neq a_3$ $\alpha = \beta = \gamma = 90^\circ$
Tetragonal	2	$a_1 = a_2 \neq a_3$ $\alpha = \beta = \gamma = 90^\circ$
Cubic	3	$a_1 = a_2 = a_3$ $\alpha = \beta = \gamma = 90^\circ$
Trigonal	1	$a_1 = a_2 = a_3$ $\alpha = \beta = \gamma < 120^\circ, \neq 90^\circ$
Hexagonal	1	$a_1 = a_2 \neq a_3$ $\alpha = \beta = 90^\circ$ $\gamma = 120^\circ$

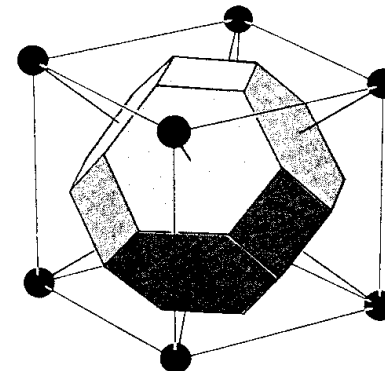
Wigner-Seitz Primitive Cell: Full symmetry of Bravais Lattice



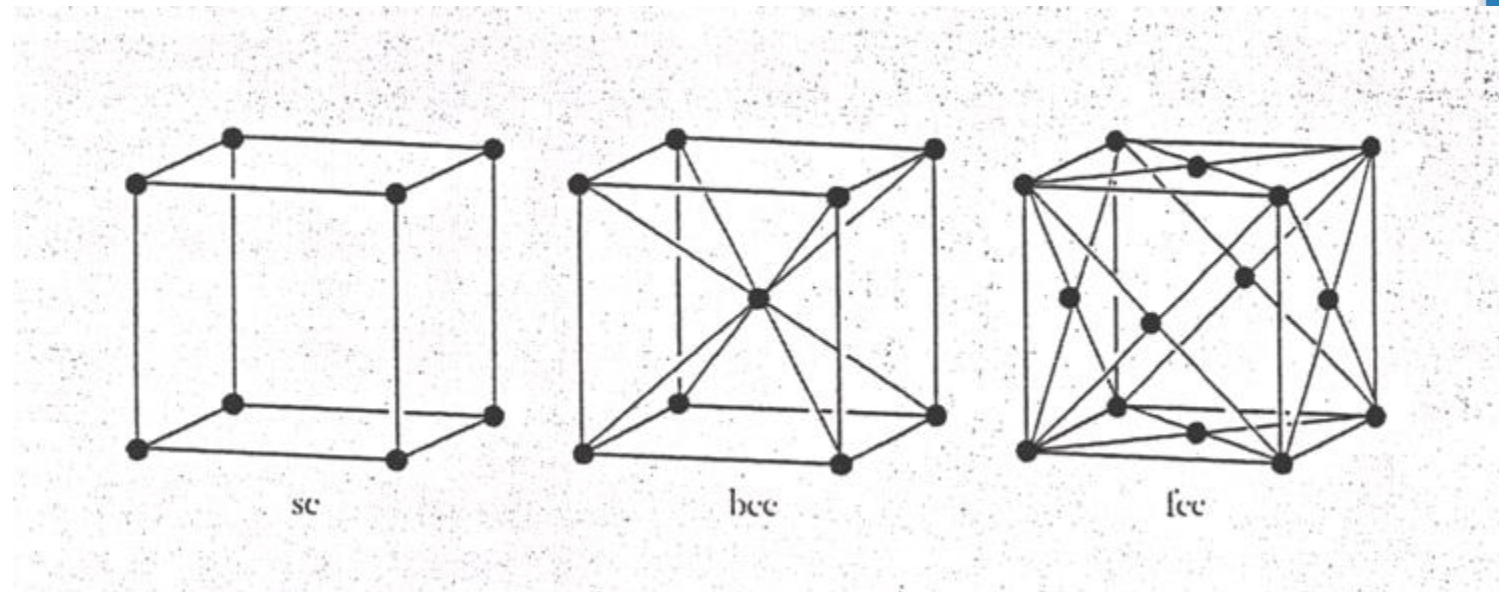
- Figure 4.14**
 The Wigner-Seitz cell for a two-dimensional Bravais lattice. The six sides of the cell bisect the lines joining the central points to its six nearest neighboring points (shown as dashed lines). In two dimensions the Wigner-Seitz cell is always a hexagon unless the lattice is rectangular (see Problem 4a).

Figure 4.15

The Wigner-Seitz cell for the body-centered cubic Bravais lattice (a “truncated octahedron”). The surrounding cube is a conventional body-centered cubic cell with a lattice point at its center and on each vertex. The hexagonal faces bisect the lines joining the central point to the points on the vertices (drawn as solid lines). The square faces bisect the lines joining the central point to the central points in each of the six neighboring cubic cells (not drawn). The hexagons are regular (see Problem 4d).



Cubic space lattices



Cubic lattices

Table 2 Characteristics of cubic lattices^a

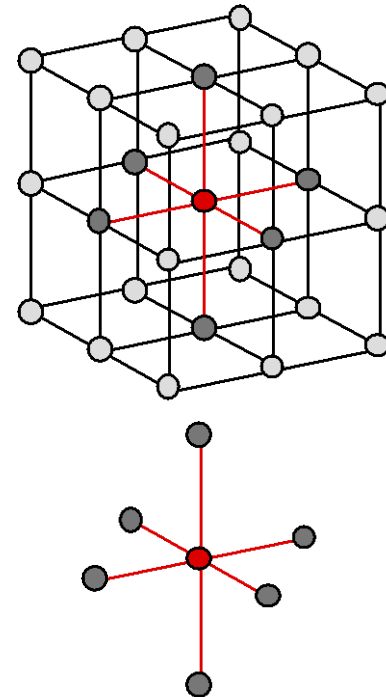
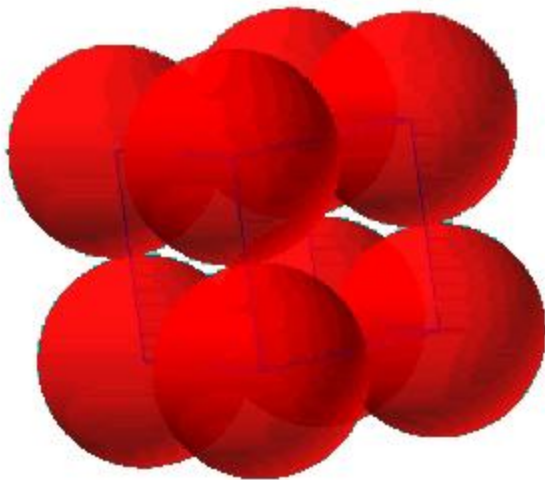
	Simple	Body-centered	Face-centered
Volume, conventional cell	a^3	a^3	a^3
Lattice points per cell	1	2	4
Volume, primitive cell	a^3	$\frac{1}{2}a^3$	$\frac{1}{4}a^3$
Lattice points per unit volume	$1/a^3$	$2/a^3$	$4/a^3$
Number of nearest neighbors ^a	6	8	12
Nearest-neighbor distance	a	$3^{1/2}a/2 = 0.866a$	$a/2^{1/2} = 0.707a$
Number of second neighbors	12	6	6
Second neighbor distance	$2^{1/2}a$	a	a
Packing fraction ^b	$\frac{1}{6}\pi$ $= 0.524$	$\frac{1}{8}\pi\sqrt{3}$ $= 0.680$	$\frac{1}{6}\pi\sqrt{2}$ $= 0.740$

^aTables of numbers of neighbors and distances in sc, bcc, fcc, hcp, and diamond structures are given on pp. 1037–1039 of J. Hirschfelder, C. F. Curtis and R. B. Bird, *Molecular theory of gases and liquids*, Wiley, 1964.

^bThe packing fraction is the maximum proportion of the available volume that can be filled with hard spheres.

SIMPLE CUBIC STRUCTURE (SC)

- Rare due to poor packing (only Po has this structure)
- **Close-packed directions** are cube edges.
- **Coordination # = 6**
(# nearest neighbors)

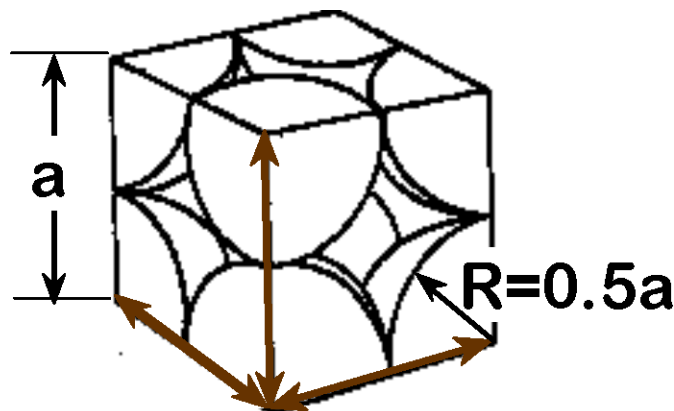


ATOMIC PACKING FACTOR

$$APF = \frac{\text{Volume of atoms in unit cell}^*}{\text{Volume of unit cell}}$$

*assume hard spheres

- APF for a simple cubic structure = 0.52



close-packed directions

contains $8 \times 1/8 =$

1 atom/unit cell

$$APF = \frac{\text{atoms unit cell} \times \frac{4}{3} \pi (0.5a)^3}{a^3}$$

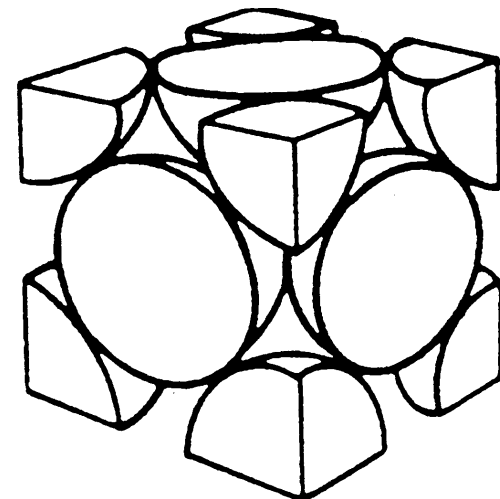
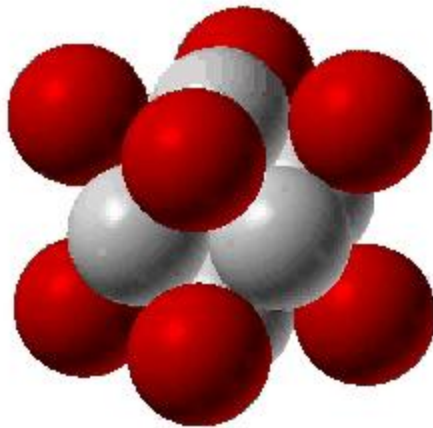
volume atom

volume unit cell

FACE CENTERED CUBIC STRUCTURE (FCC)

- Close packed directions are face diagonals.
--Note: All atoms are identical; the face-centered atoms are shaded differently only for ease of viewing.

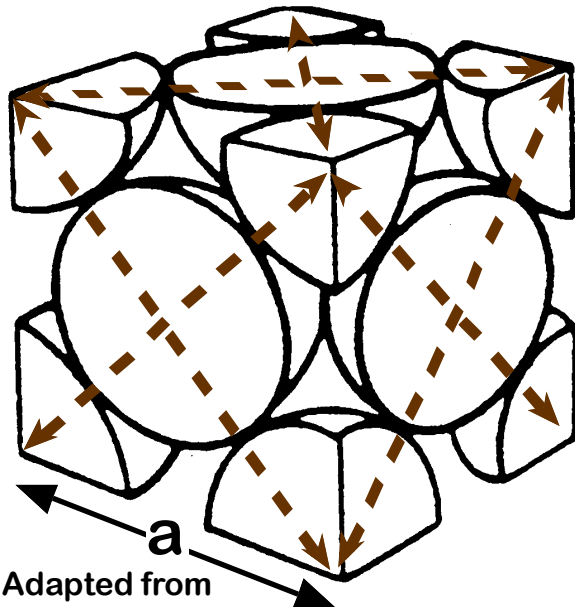
- Coordination # = 12



Adapted from Fig. 3.1(a),
Callister 6e.

ATOMIC PACKING FACTOR: FCC

- APF for a body-centered cubic structure = 0.74



Adapted from
Fig. 3.1(a),
Callister 6e.

Close-packed directions:
length = $4R$
 $= \sqrt{2} a$

Unit cell contains:
 $6 \times 1/2 + 8 \times 1/8$
 $= 4 \text{ atoms/unit cell}$

$$\text{APF} = \frac{\text{atoms unit cell} \times \frac{4}{3} \pi (\sqrt{2}a/4)^3}{a^3}$$

$\xleftarrow{\text{volume atom}}$
 $\xleftarrow{\text{volume unit cell}}$

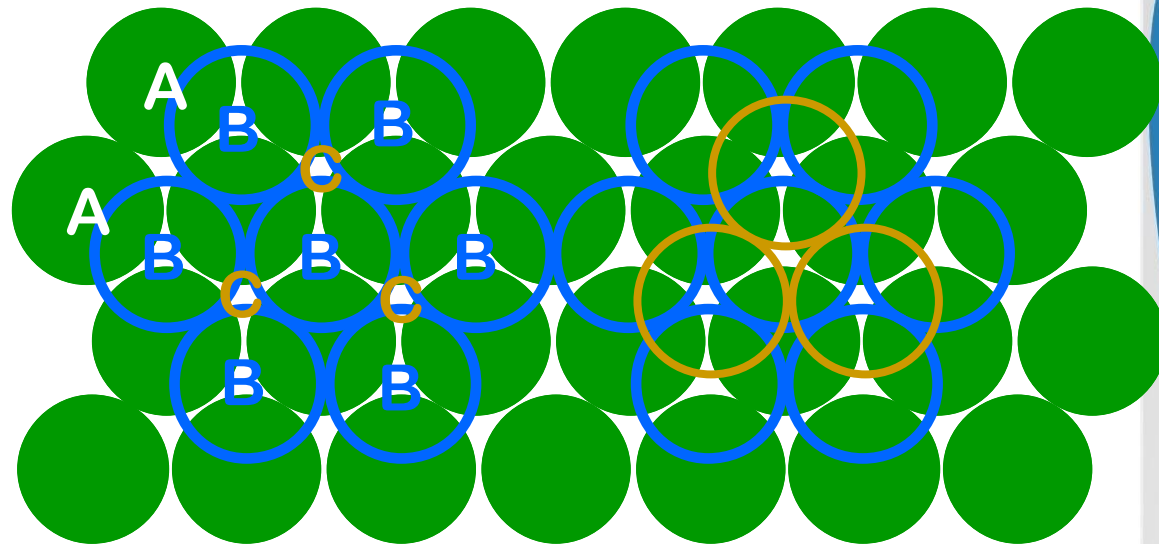
FCC STACKING SEQUENCE

- ABCABC... Stacking Sequence
- 2D Projection

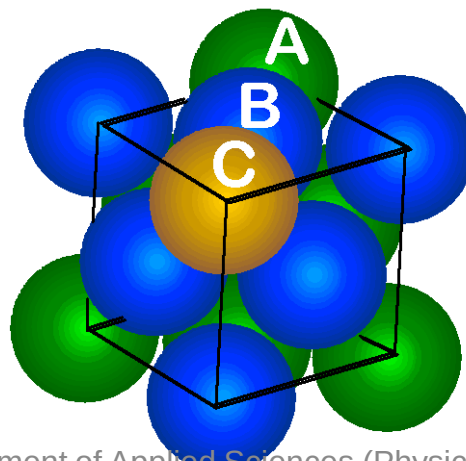
A sites

B sites

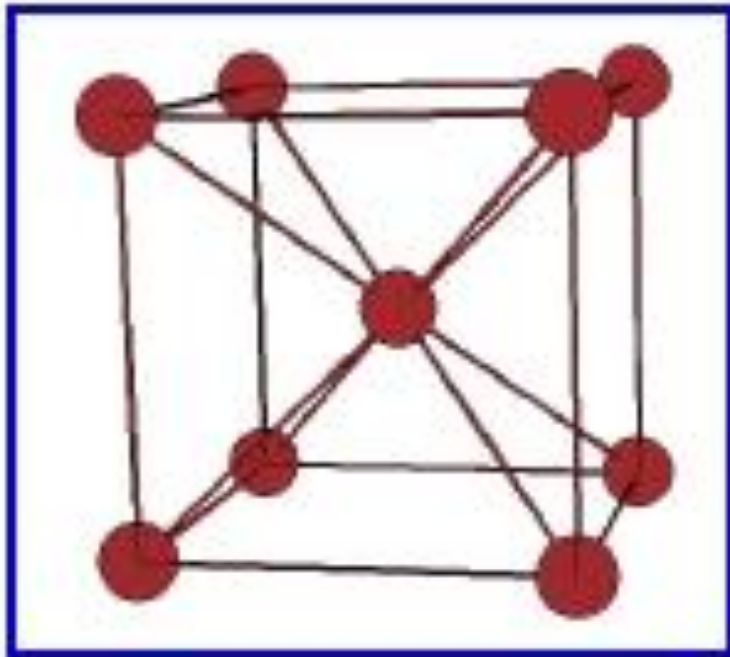
C sites



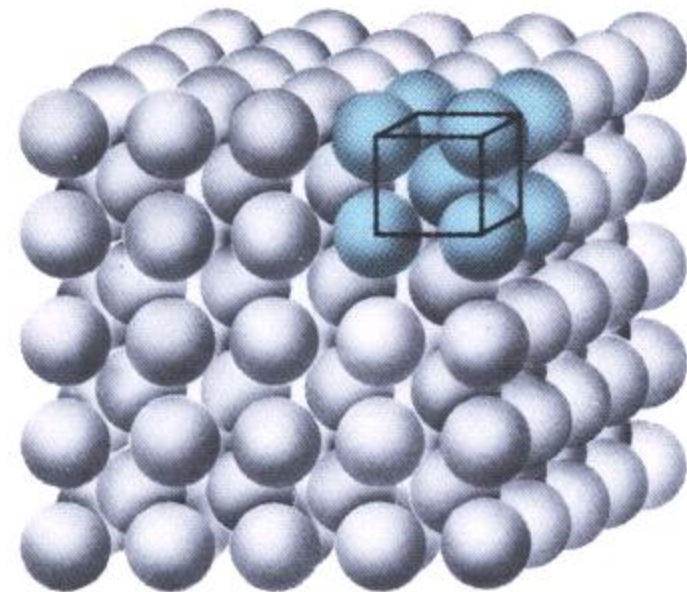
- FCC Unit Cell



BCC Structure

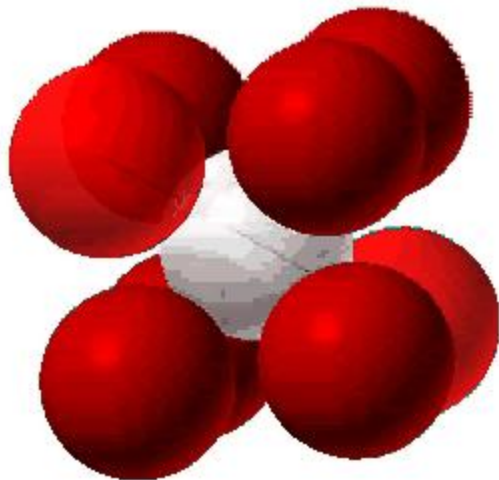


Body Centered Cubic
and related structures

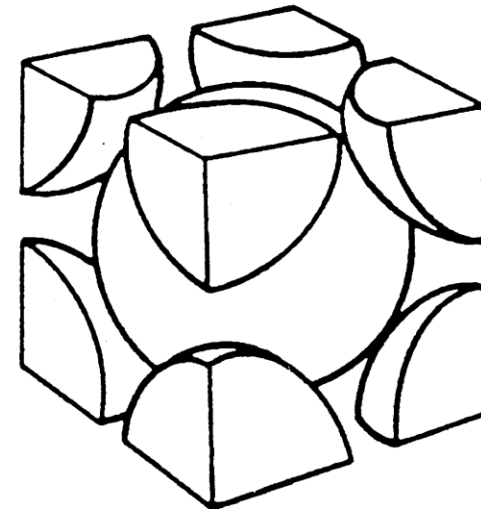


BODY CENTERED CUBIC STRUCTURE (BCC)

- Close packed directions are cube diagonals.
--Note: All atoms are identical; the center atom is shaded differently only for ease of viewing.



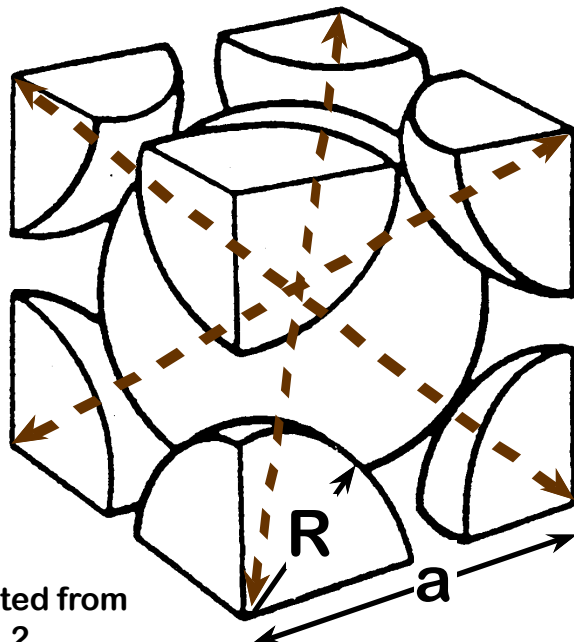
- Coordination # = 8



Adapted from Fig. 3.2,
Callister 6e.

ATOMIC PACKING FACTOR: BCC

- APF for a body-centered cubic structure = 0.68



Adapted from
Fig. 3.2,
Callister 6e.

Close-packed directions:
length = $4R$
 $= \sqrt{3} a$

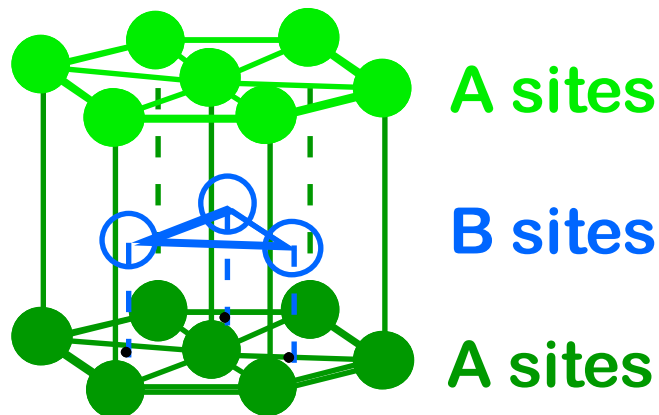
Unit cell contains:
 $1 + 8 \times 1/8$
 $= 2 \text{ atoms/unit cell}$

$$\text{APF} = \frac{\text{atoms unit cell} \times \frac{4}{3} \pi (\sqrt{3}a/4)^3}{a^3}$$

atoms unit cell → 2 $\frac{4}{3} \pi (\sqrt{3}a/4)^3$ ← volume atom
 APF = a^3 ← volume unit cell

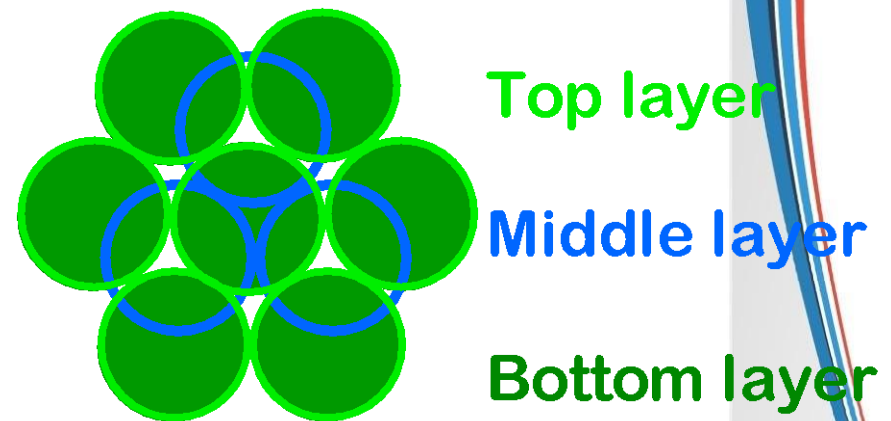
HEXAGONAL CLOSE-PACKED STRUCTURE (HCP)

- ABAB... Stacking Sequence
- 3D Projection



Adapted from Fig. 3.3,
Callister 6e.

- 2D Projection



- Coordination # = 12
- APF = 0.74

Primitive vectors BCC

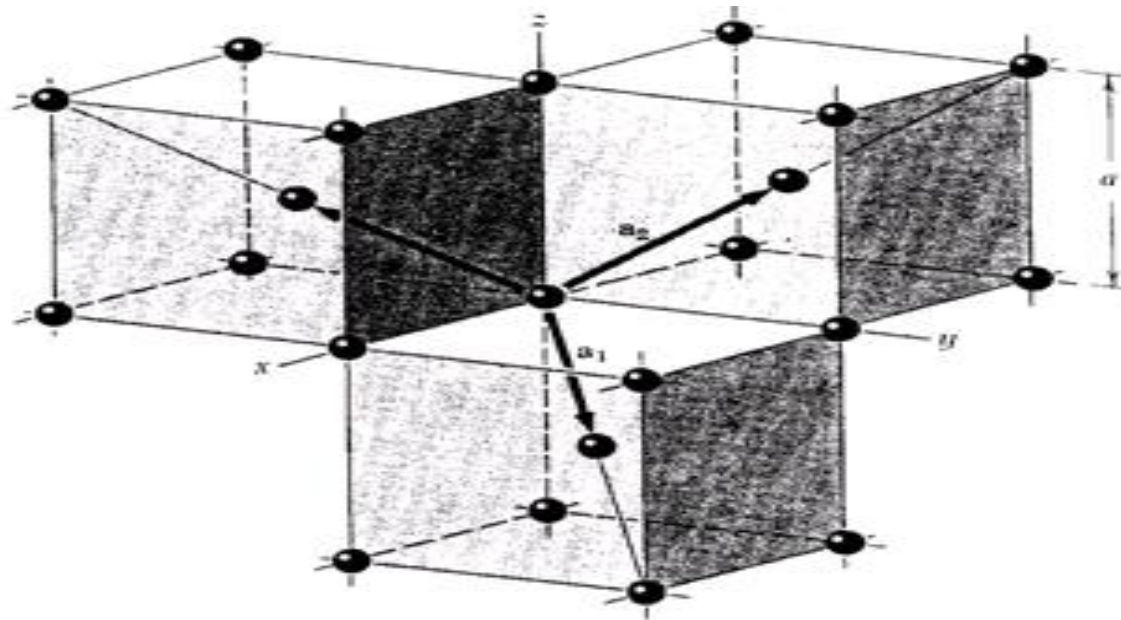


Figure 12 Primitive translation vectors of the body-centered cubic lattice; these vectors connect the lattice point at the origin to lattice points at the body centers. The primitive cell is obtained on completing the rhombohedron. In terms of the cube edge a the primitive translation vectors are

$$\begin{aligned} \mathbf{a}_1 &= \frac{1}{2}a(\hat{x} + \hat{y} - \hat{z}) ; & \mathbf{a}_2 &= \frac{1}{2}a(-\hat{x} + \hat{y} + \hat{z}) ; \\ \mathbf{a}_3 &= \frac{1}{2}a(\hat{x} - \hat{y} + \hat{z}) . \end{aligned}$$

Elements with BCC Structure

Table 4.2

ELEMENTS WITH THE MONATOMIC BODY-CENTERED CUBIC CRYSTAL STRUCTURE

ELEMENT	a (Å)	ELEMENT	a (Å)	ELEMENT	a (Å)
Ba	5.02	Li	3.49 (78 K)	Ta	3.31
Cr	2.88	Mo	3.15	Tl	3.88
Cs	6.05 (78 K)	Na	4.23 (5 K)	V	3.02
Fe	2.87	Nb	3.30	W	3.16
K	5.23 (5 K)	Rb	5.59 (5 K)		

Summary: Bravais Lattices (Nets) in Two Dimensions

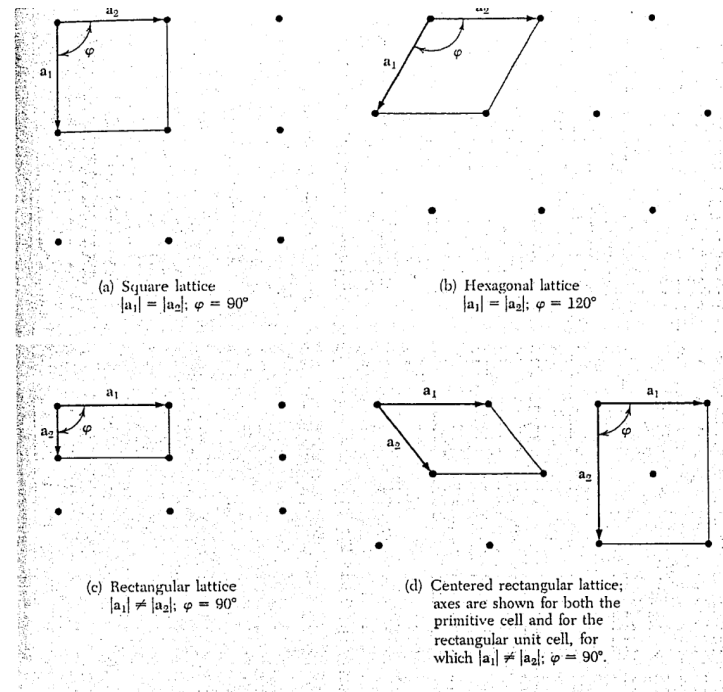


Figure 9

Summary: Fourteen Bravais Lattices in Three Dimensions

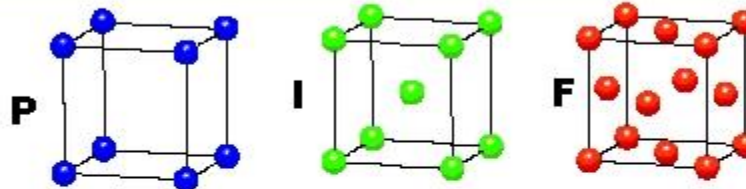
The 14 possible **BRAVAIS LATTICES**

{note that spheres in this picture represent lattice points, not atoms!}

CUBIC

$$a = b = c$$

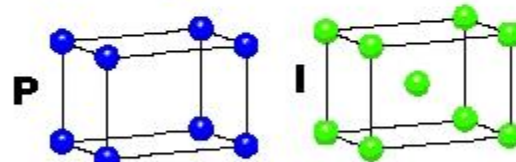
$$\alpha = \beta = \gamma = 90^\circ$$



TETRAGONAL

$$a = b \neq c$$

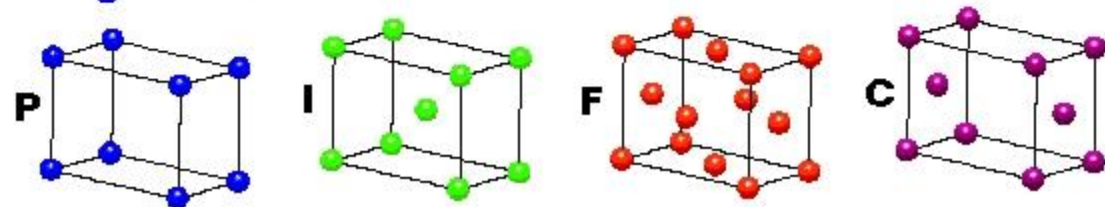
$$\alpha = \beta = \gamma = 90^\circ$$



ORTHORHOMBIC

$$a \neq b \neq c$$

$$\alpha = \beta = \gamma = 90^\circ$$



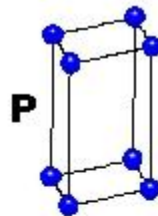
Fourteen Bravais Lattices ...

HEXAGONAL

$$a = b \neq c$$

$$\alpha = \beta = 90^\circ$$

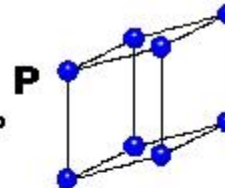
$$\gamma = 120^\circ$$



TRIGONAL

$$a = b = c$$

$$\alpha = \beta = \gamma \neq 90^\circ$$

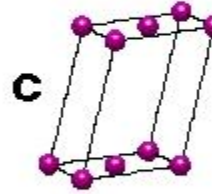
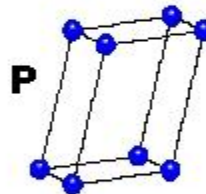


MONOCLINIC

$$a \neq b \neq c$$

$$\alpha = \gamma = 90^\circ$$

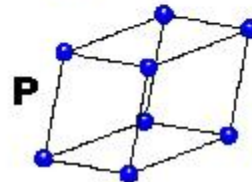
$$\beta \neq 120^\circ$$



TRICLINIC

$$a \neq b \neq c$$

$$\alpha \neq \beta \neq \gamma \neq 90^\circ$$



4 Types of Unit Cell

P = Primitive

I = Body-Centred

F = Face-Centred

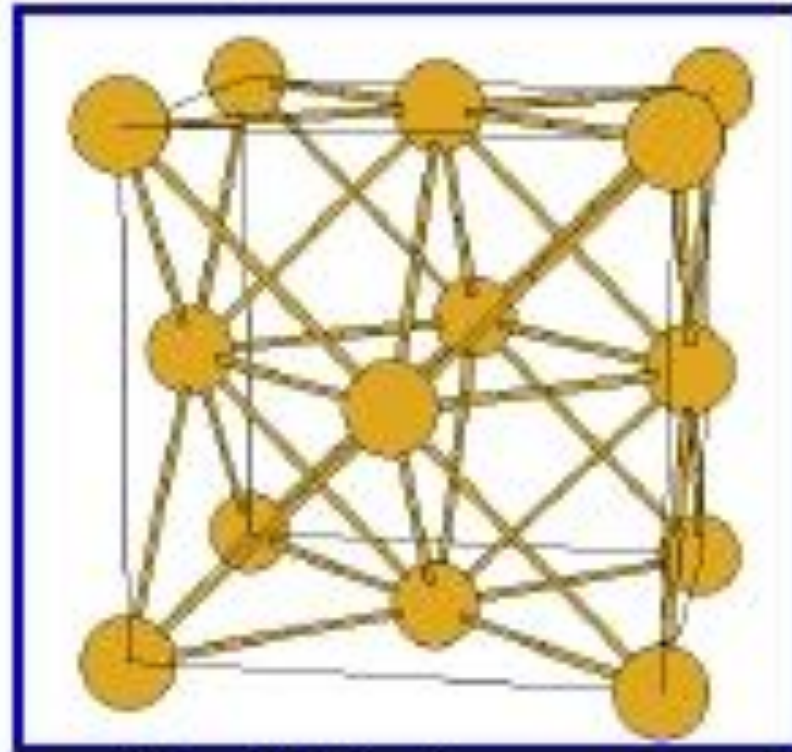
C = Side-Centred

+

7 Crystal Classes

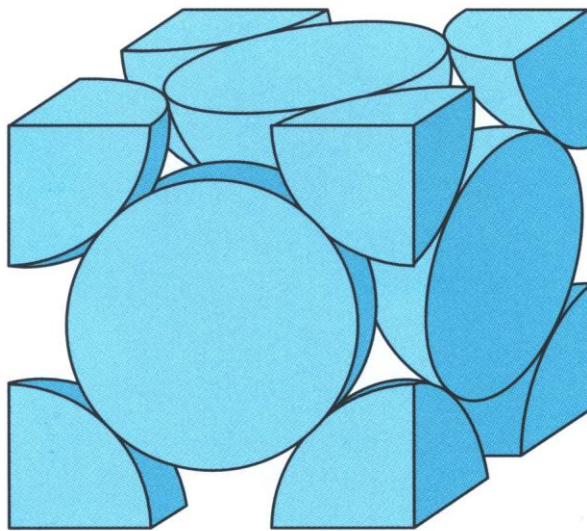
→ 14 Bravais Lattices

FCC Structure

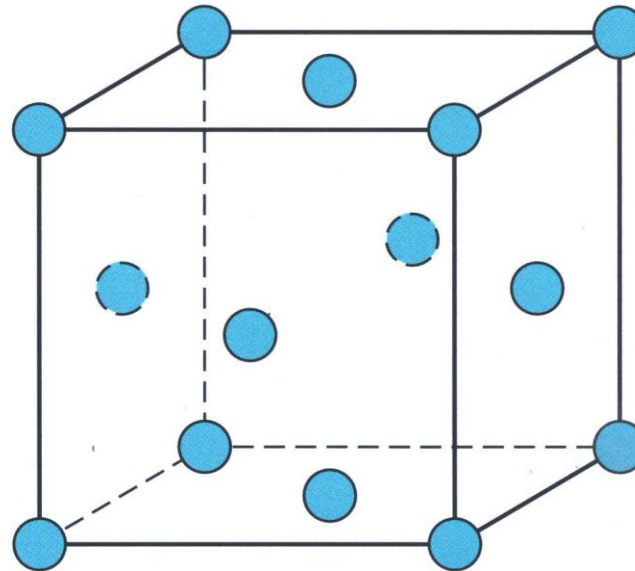


A1 (fcc) Structure

FCC lattice



(a)



(b)

Primitive Cell: FCC Lattice

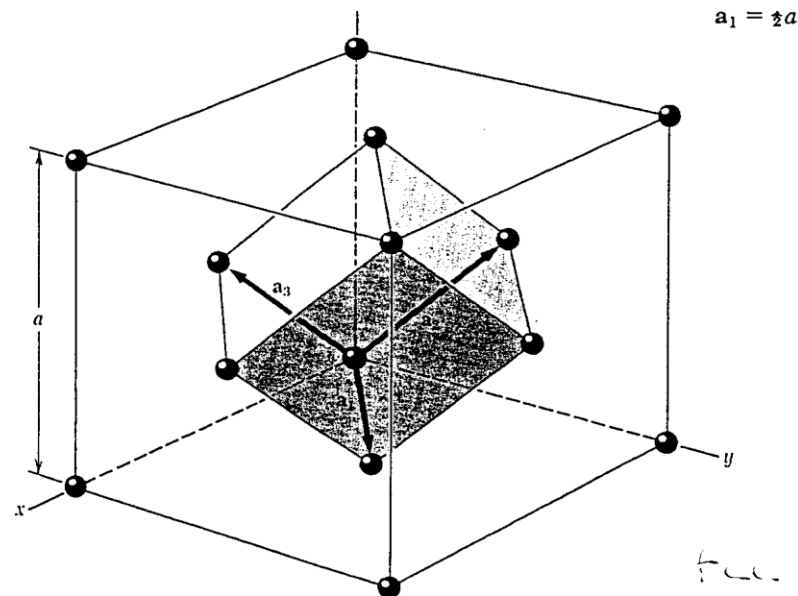


Figure 13 The rhombohedral primitive cell of the face-centered cubic crystal. The primitive translation vectors a_1 , a_2 , a_3 connect the lattice point at the origin with lattice points at the face centers. As drawn, the primitive vectors are:

$$a_1 = \frac{1}{2}a(\hat{x} + \hat{y}) ; \quad a_2 = \frac{1}{2}a(\hat{y} + \hat{z}) ; \quad a_3 = \frac{1}{2}a(\hat{z} + \hat{x}) .$$

The angles between the axes are 60° . Here $\hat{x}$, $\hat{y}$, $\hat{z}$ are the Cartesian unit vectors.

FCC: Conventional Cell With Basis

- We can also view the FCC lattice in terms of a conventional unit cell with a four point basis.
- Similarly, we can view the BCC lattice in terms of a conventional unit cell with a two point basis.

Elements That Have FCC Structure

ELEMENTS WITH THE MONATOMIC FACE-CENTERED CUBIC CRYSTAL STRUCTURE

ELEMENT	a (Å)	ELEMENT	a (Å)	ELEMENT	a (Å)
Ar	5.26 (4.2 K)	Ir	3.84	Pt	3.92
Ag	4.09	Kr	5.72 (58 K)	δ -Pu	4.64
Al	4.05	La	5.30	Rh	3.80
Au	4.08	Ne	4.43 (4.2 K)	Sc	4.54
Ca	5.58	Ni	3.52	Sr	6.08
Ce	5.16	Pb	4.95	Th	5.08
β -Co	3.55	Pd	3.89	Xe (58 K)	6.20
Cu	3.61	Pr	5.16	Yb	5.49

Data in Tables 4.1 to 4.7 are from R. W. G. Wyckoff, *Crystal Structures*, 2nd ed., Interscience, New York, 1963. In most cases, the data are taken at about room temperature and normal atmospheric pressure. For elements that exist in many forms the stable room temperature form (or forms) is given. For more detailed information, more precise lattice constants, and references, the Wyckoff work should be consulted.

Simple Hexagonal Bravais Lattice

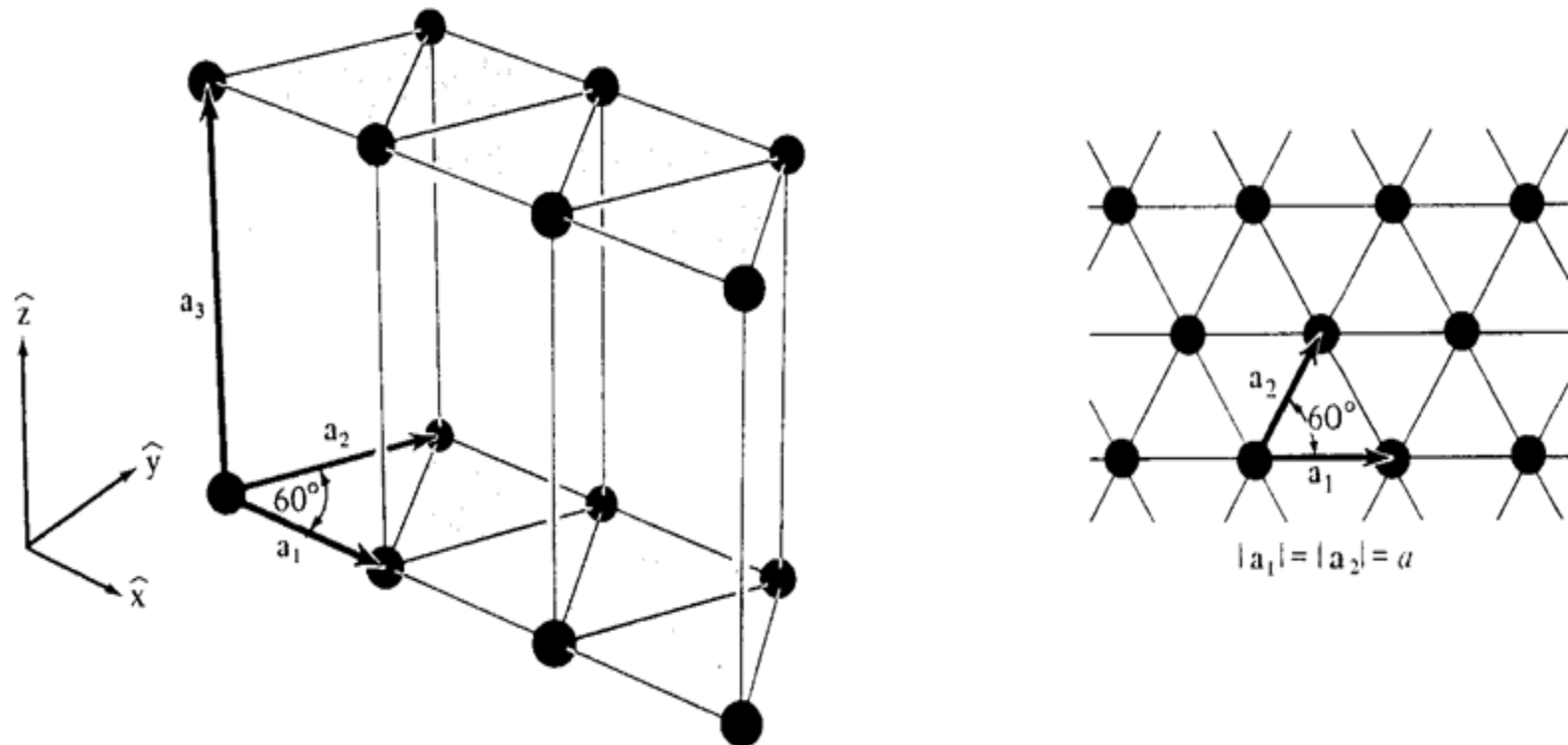


Figure 4.19

The simple hexagonal Bravais lattice. Two-dimensional triangular nets (shown in inset) are stacked directly above one another, a distance c apart.

Primitive Cell: Hexagonal System

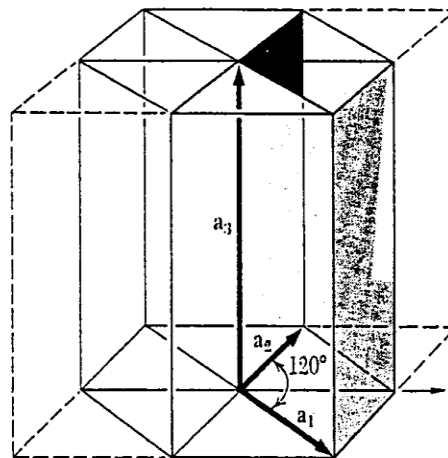
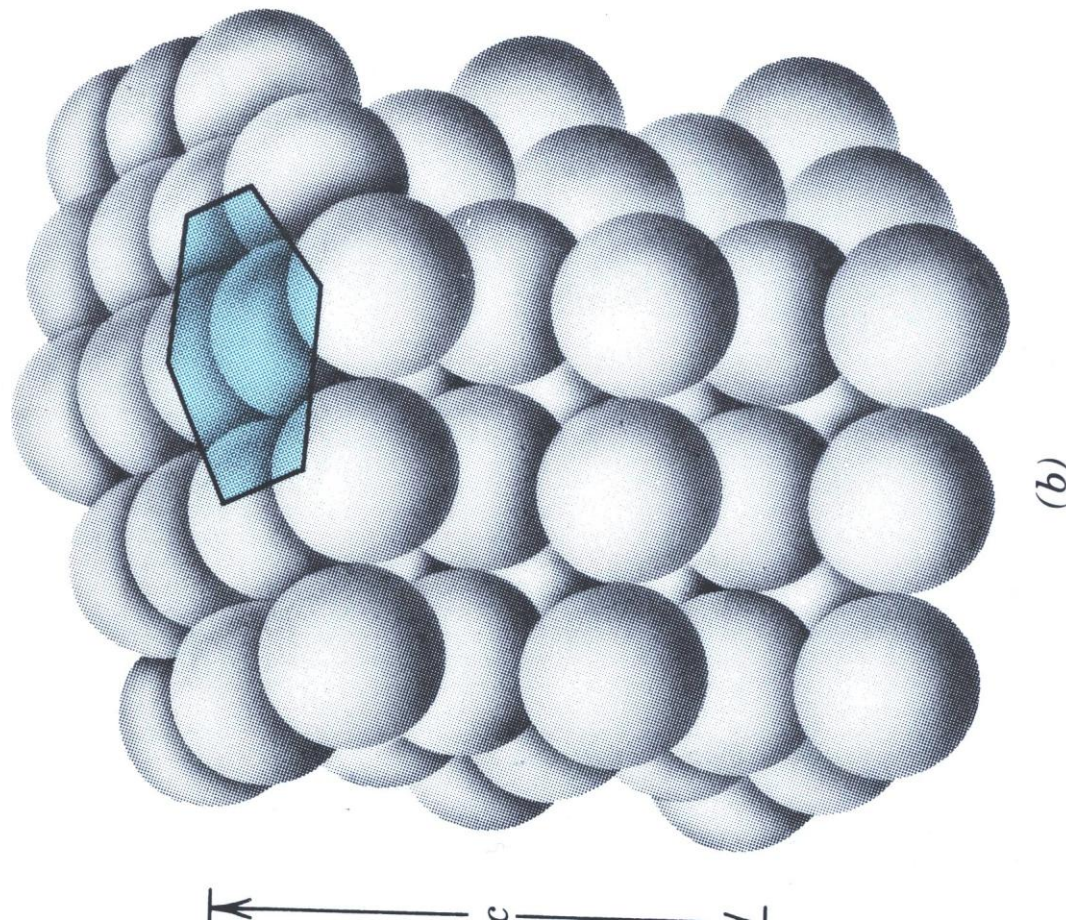


Figure 14 Relation of the primitive cell in the hexagonal system (heavy lines) to a prism of hexagonal symmetry. Here $a_1 = a_2 \neq a_3$.

HCP Crystal



Hexagonal Close Packing

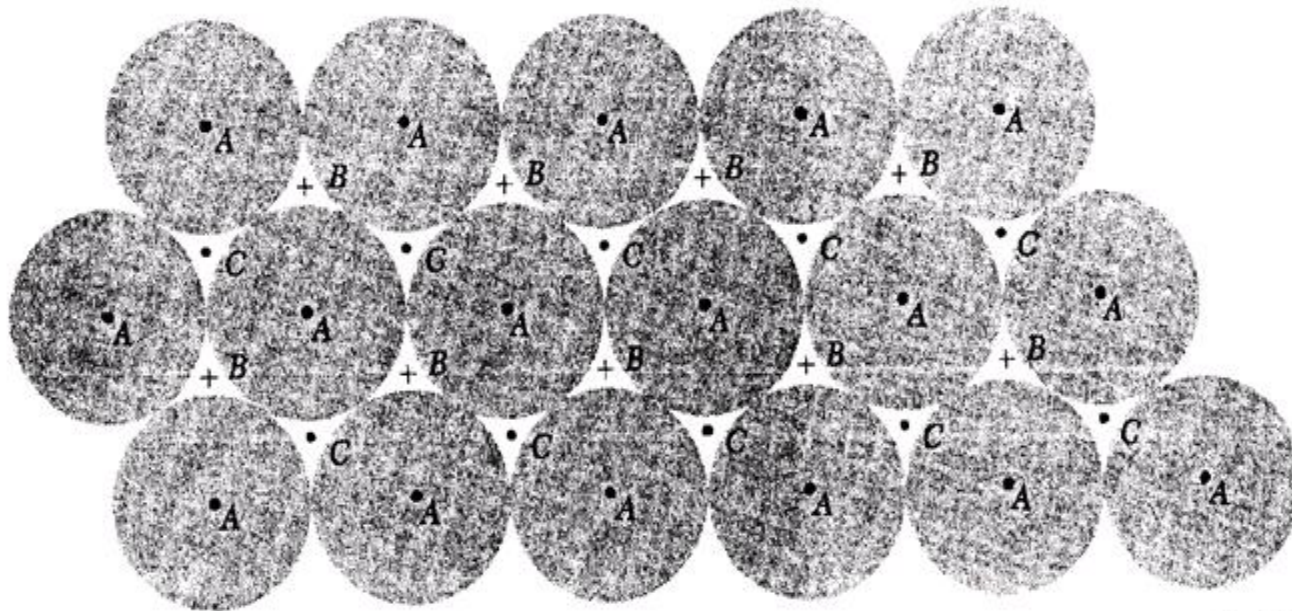
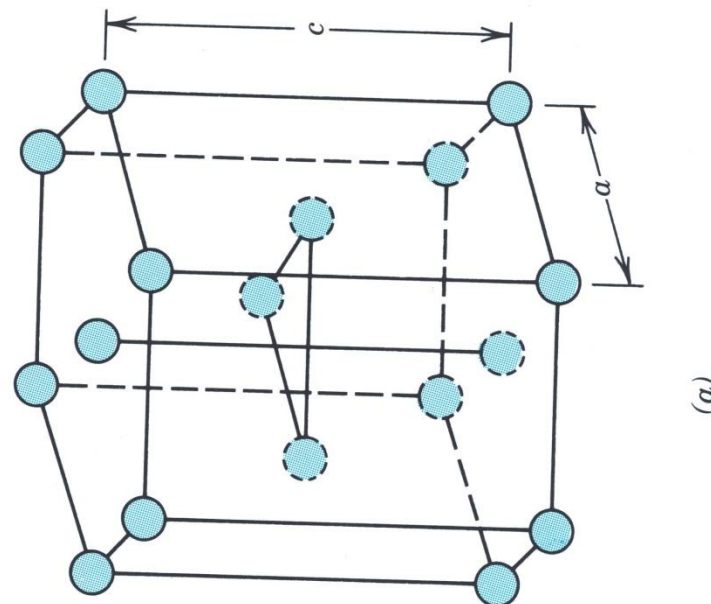


Figure 21 A close-packed layer of spheres is shown, with centers at points marked A. A second and identical layer of spheres can be placed on top of this, above and parallel to the plane of the drawing, with centers over the points marked B. There are two choices for a third layer. It can go in over A or over C. If it goes in over A the sequence is $ABABAB \dots$ and the structure is hexagonal close-packed. If the third layer goes in over C the sequence is $ABCABCABC \dots$ and the structure is face-centered cubic.

HexagonalClosePacked



HCP lattice is not a Bravais lattice, because orientation of the environment Of a point varies from layer to layer along the c-axis.

Miller indices of lattice plane

- The indices of a crystal plane (h,k,l) are defined to be a set of **integers** with no common factors, inversely proportional to the intercepts of the crystal plane along the crystal axes:

$$h:k:l = \frac{1}{x_1} : \frac{1}{x_2} : \frac{1}{x_3}$$

Indices of Crystal Plane

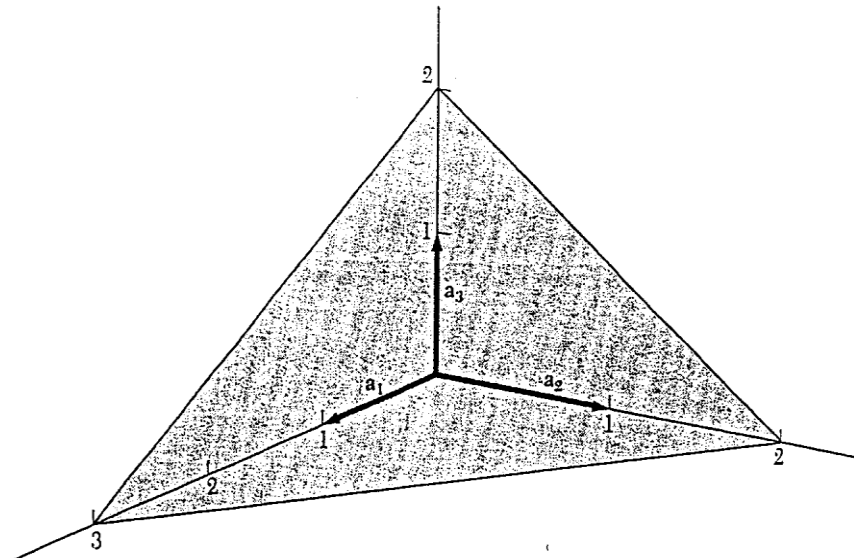


Figure 15 This plane intercepts the a_1 , a_2 , a_3 axes at $3a_1$, $2a_2$, $2a_3$. The reciprocals of these numbers are $\frac{1}{3}$, $\frac{1}{2}$, $\frac{1}{2}$. The smallest three integers having the same ratio are 2, 3, 3, and thus the indices of the plane are (233).

Indices of Planes: Cubic Crystal

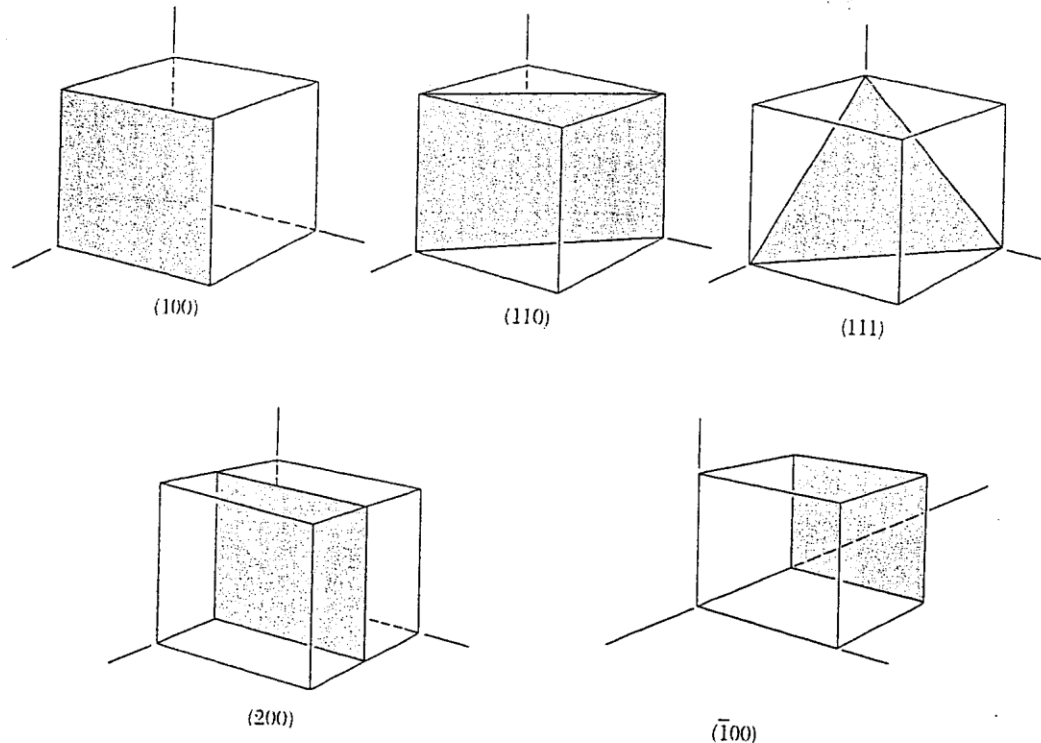
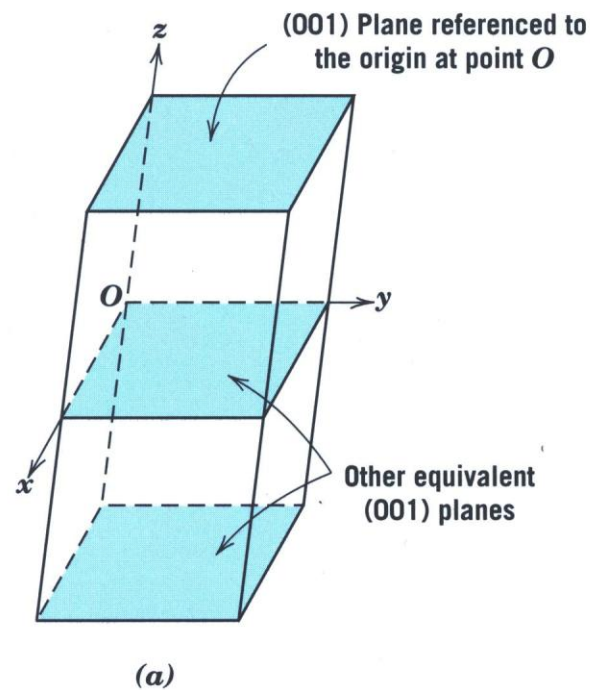


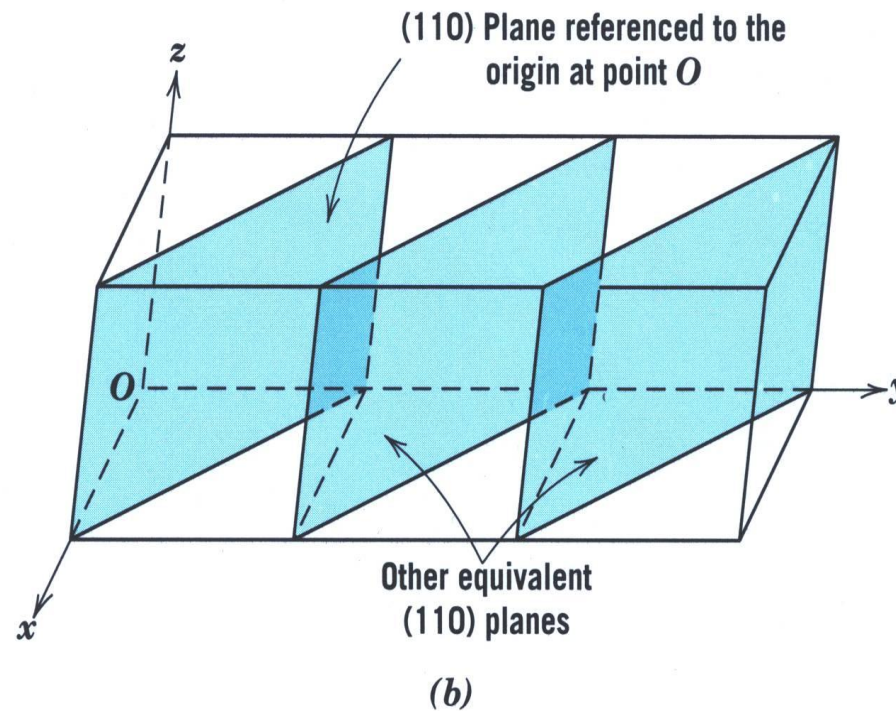
Figure 16 Indices of important planes in a cubic crystal. The plane (200) is parallel to (100) and to $(\bar{1}00)$.

001 Plane

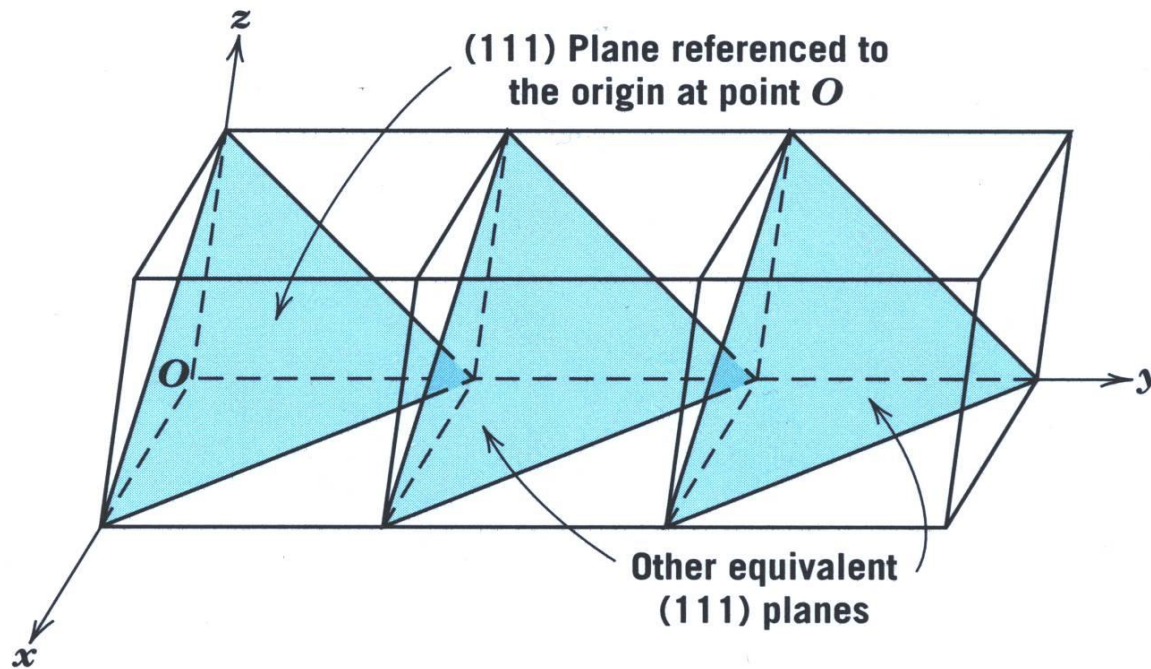


x'

110 Planes



111 Planes

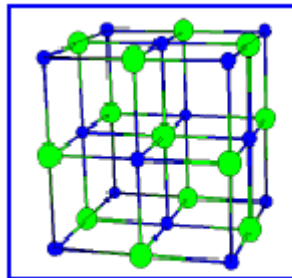


(c)

Simple Crystal Structures

- There are several crystal structures of common interest: sodium chloride, cesium chloride, hexagonal close-packed, diamond and cubic zinc sulfide.
- Each of these structures have many different realizations.

NaCl Structure



1 (Sodium Chloride) Structure

NaCl Basis

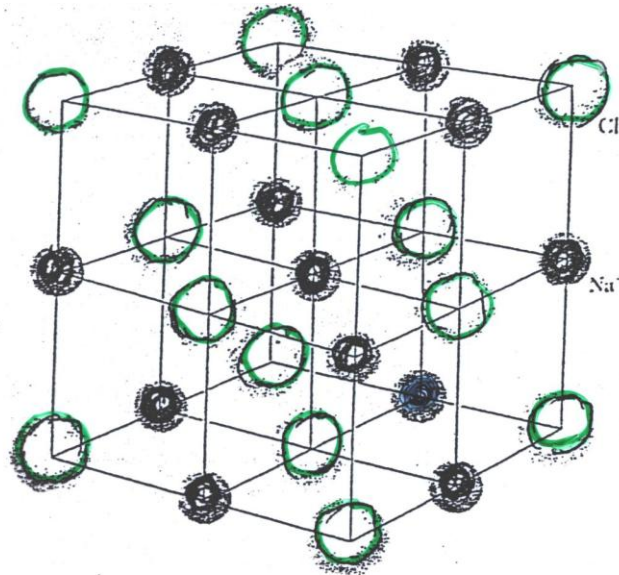
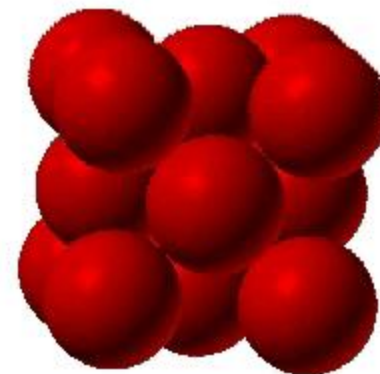
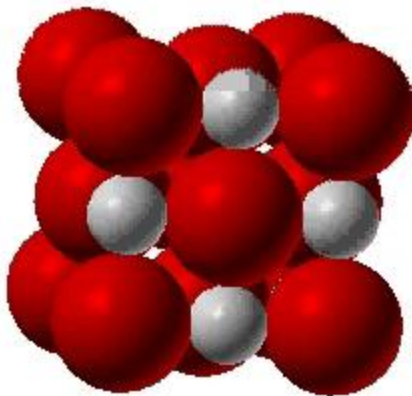


Figure 17 We may construct the sodium chloride crystal structure by arranging Na^+ and Cl^- ions alternately at the lattice points of a simple cubic lattice. In the crystal each ion is surrounded by six nearest neighbors of the opposite charge. The space lattice is fcc, and the basis has one Cl^- ion at 000 and one Na^+ ion at $\frac{1}{2}\frac{1}{2}\frac{1}{2}$. The figure shows one conventional cubic cell. The ionic diameters here are reduced in relation to the cell in order to clarify the spatial arrangement.

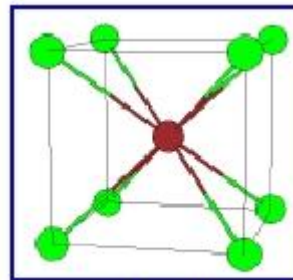
STRUCTURE OF COMPOUNDS

NaCl

- Compounds: Often have similar close-packed structures.
- Structure of NaCl
- Close-packed directions --along cube edges.



CsCl Structure



B2 (CsCl) Structure

CsCl Basis

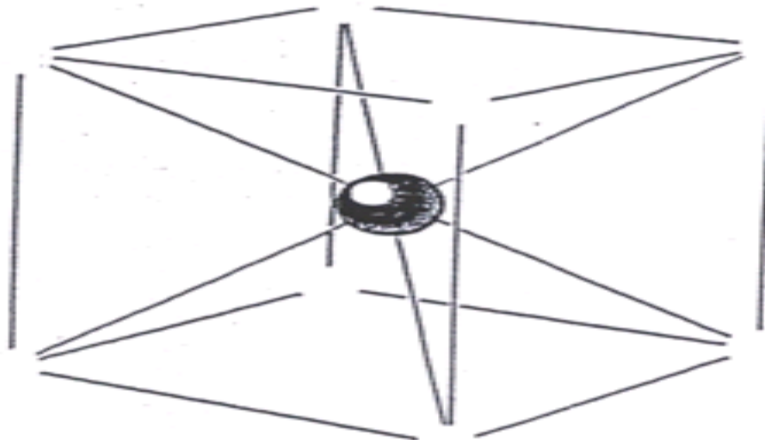
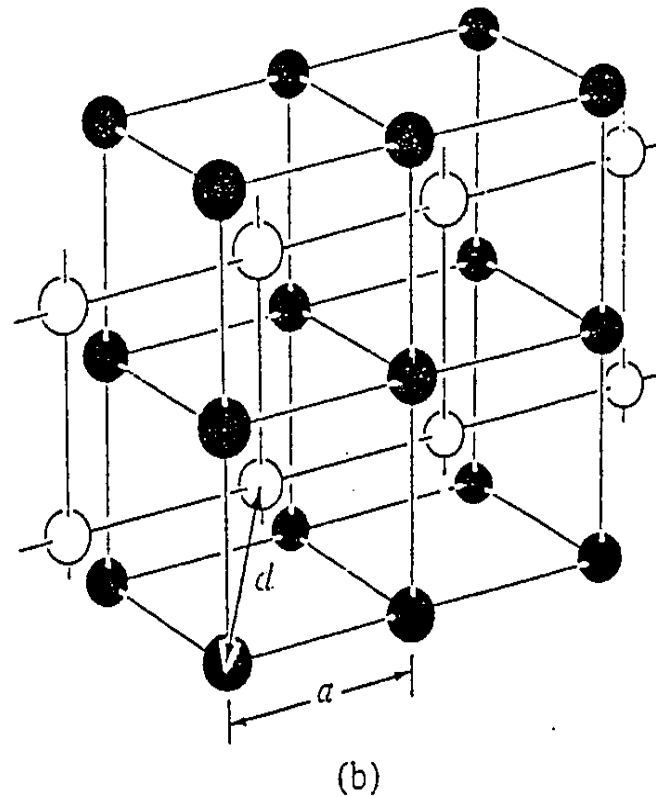


Figure 20 The cesium chloride crystal structure. The space lattice is simple cubic, and the basis has one Cs^+ ion at 000 and one Cl^- ion at $\frac{1}{2}\frac{1}{2}\frac{1}{2}$.

CsCl Basis



Diamond Crystal Structure

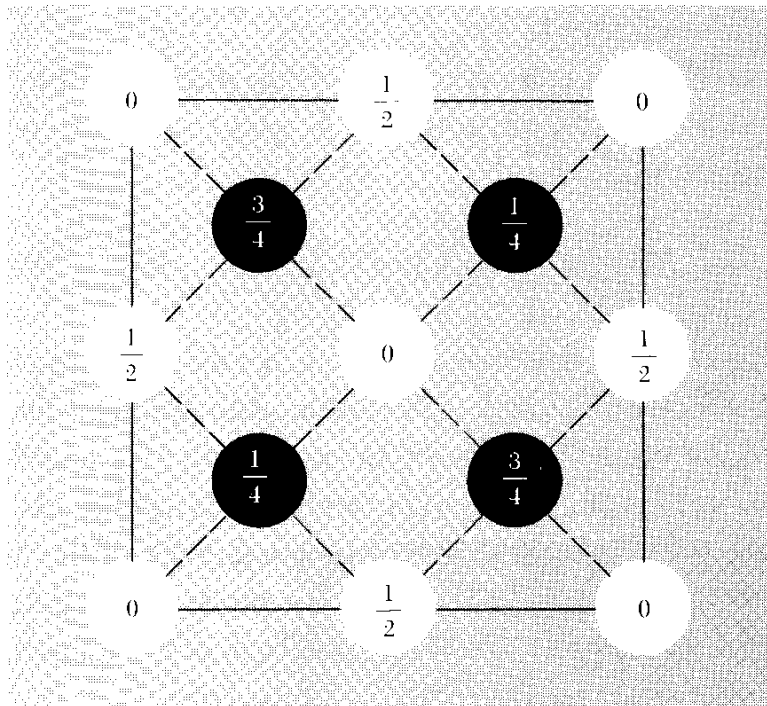


Figure 24 Atomic positions in the cubic cell of the diamond

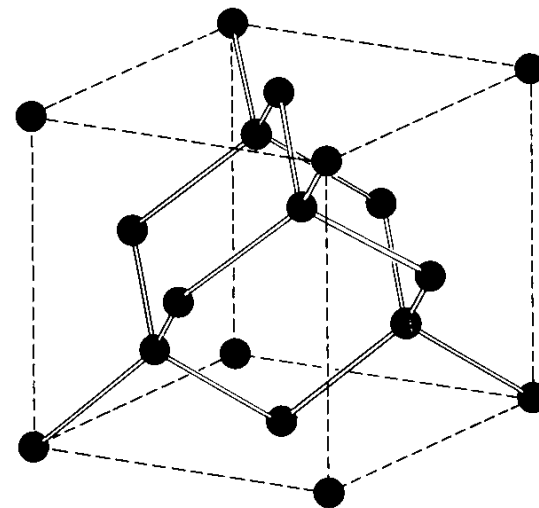


Figure 25 Crystal structure of diamond,

Symmetry planes

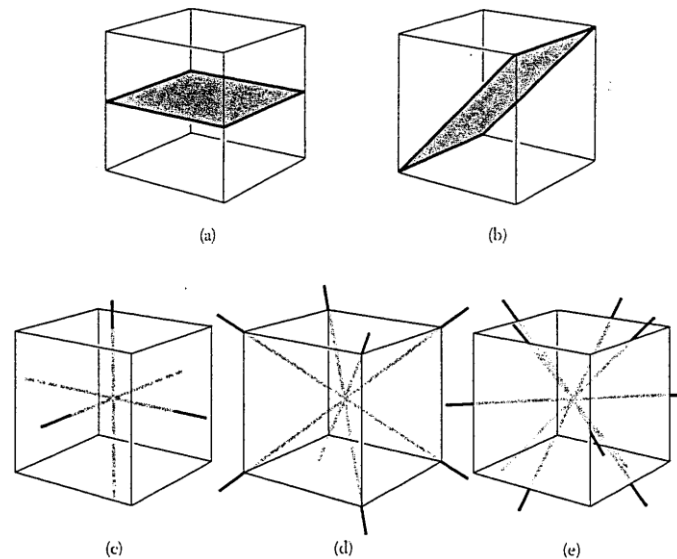
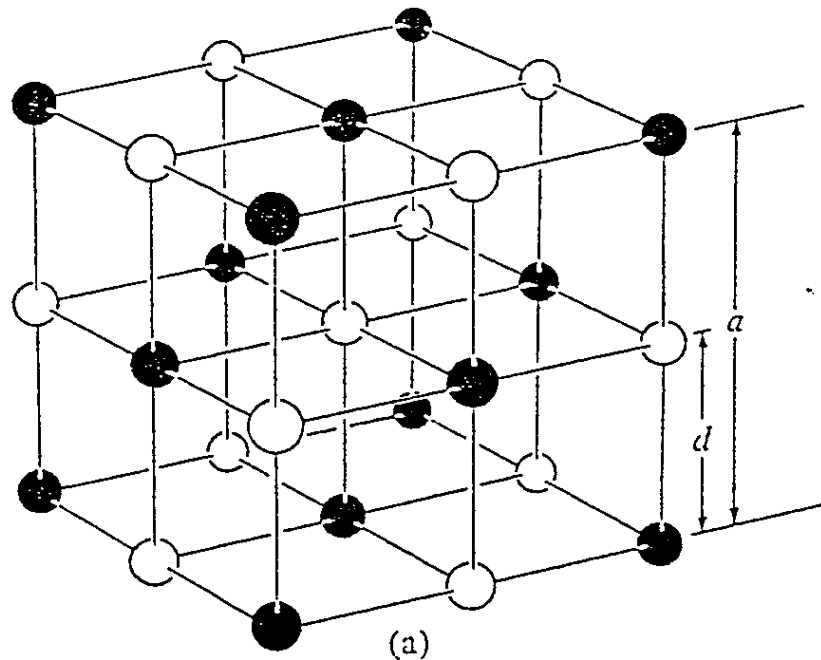


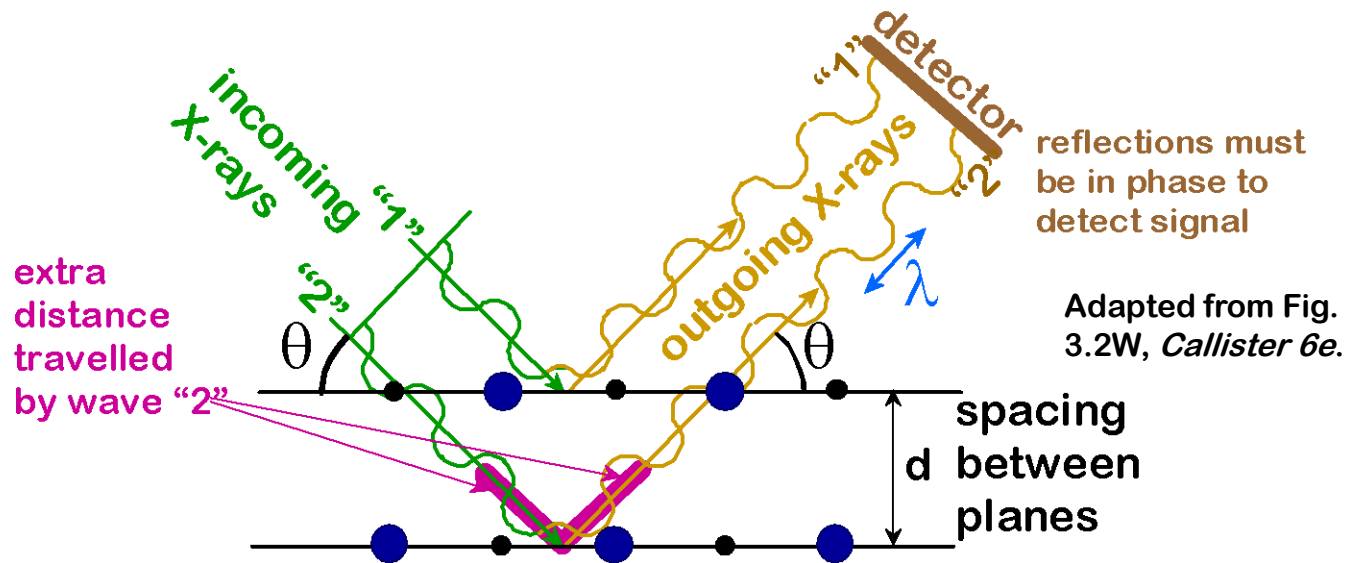
Figure 8 (a) A plane of symmetry parallel to the faces of a cube. (b) A diagonal plane of symmetry in a cube. (c) The three tetrad axes of a cube. (d) The four triad axes of a cube. (e) The six diad axes of a cube.

NaCl Basis



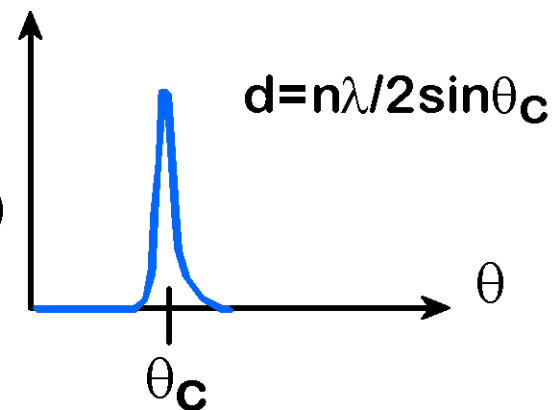
X-RAYS TO CONFIRM CRYSTAL STRUCTURE

- Incoming X-rays **diffract** from crystal planes.



- Measurement of:
Critical angles, θ_c ,
for X-rays provide
atomic spacing, d .

x-ray
intensity
(from
detector)



SUMMARY

- Atoms may assemble into **crystalline** or **amorphous** structures.
- We can predict the **density** of a material, provided we know the **atomic weight**, **atomic radius**, and **crystal geometry** (e.g., FCC, BCC, HCP).
- Material properties generally vary with single crystal orientation (i.e., they are **anisotropic**), but properties are generally non-directional (i.e., they are **isotropic**) in polycrystals with randomly oriented grains.